SULTAN QABOOS UNIVERSITY

College of Engineering

Department of Electrical and Computer Engineering

Computer Systems and Networks

Smart Healthcare Robot

Bringing Care Closer to Patients in Oman

Final Year Project Report

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Academic Year 2025–2026

December 21, 2025

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Introduction

This chapter establishes the project context, problem statement, motivation, objectives, methodology overview, and the project management plan that frames execution for the rest of the report.

1 Introduction

Healthcare in Oman faces a simple but serious challenge: doctors cannot always be where patients need them. This affects everyone - elderly people who cannot travel to hospitals, children in rural areaswho need specialist care, and patients in remote communities where the nearest doctor might be hours away. While Oman's healthcare system reaches 100% of urban families, 10% of rural families still lack adequate access to medical care.When doctors are unavailable due to emergencies, travel, or simply being in another location, patients wait. Sometimes they wait too long. This is particularly difficult for elderly patients who may have mobility issues, and for children who need comfort and reassurance during medical situations.[1]

Oman's healthcare system strives to reach all citizens, with government local health services reaching 100% of urban families and 90% of rural families [1], yet geographic distances and resource constraints create barriers to timely and accessible care for the remaining underserved populations. Existing telepresence solutions often rely on manual control and lack the ability to engage patients in even basic diagnostic dialogue. This results in increased workloads for healthcare providers and a less personalized, less effective experience for patients, particularly those in rural areas or with mobility limitations.

Every day in Oman, patients need medical attention that could be provided remotely if the right tools existed. Children in hospitals need comfort when parents cannot be present. Elderly patients need medication reminders and someone to check on their wellbeing. Rural communities need access to specialist consultations without long journeys.

The global demand for remote healthcare solutions is growing rapidly, with the telepresence robot market expected to reach significant growth. More importantly, Oman's Vision 2040 calls for healthcare innovation that serves all citizens equally, regardless of location [2]. Recent studies have shown that telemedicine adoption in Oman's primary healthcare settings demonstrates both promise and areas for improvement [3], while infrastructure requirements for effective telemedicine networks continue to be refined [4]. The aim of the project is to design and implement a smart telepresence robot with autonomous indoor navigation and on-device basic triage dialogue for ward corridors and bedside spaces in hospitals in Oman.

- To develop a robust autonomous navigation system capable of dynamic obstacle avoidance and Simultaneous Localization and Mapping (SLAM) in crowded hospital corridors.
- To engineer a secure telepresence interface that enables real-time, low-latency audio/visual consultations for remote doctors.
- To integrate an AI-driven interaction layer (using Large Language Models) to provide basic triage, visitor guidance, and patient reassurance.
- To validate the system in a pilot scenario, demonstrating successful autonomous delivery (Nurse Station → Bed) and remote consultation cycles.

The project begins with a systematic requirements analysis informed by literature review and healthcare facility needs assessment. Following this, we evaluate and select optimal design alternatives for both the hardware platform and software architecture using Analytic Hierarchy Process (AHP) methodology. The development follows an iterative prototyping cycle, beginning with establishing reliable remote tele-operation, advancing to implementing and refining the autonomous navigation system, and culminating in integrating the conversational AI dialogue capabilities. Finally, the system undergoes real-world validation in simulated and pilot clinical settings to measure performance metrics such as navigation reliability, telepresence quality of service, and overall user experience for both patients and clinical staff.

The project timeline spans two academic semesters, from September 2025 to May 2026. The first semester focuses on research, requirements analysis, and conceptual design, while the second semester is dedicated to implementation, testing, and final evaluation.

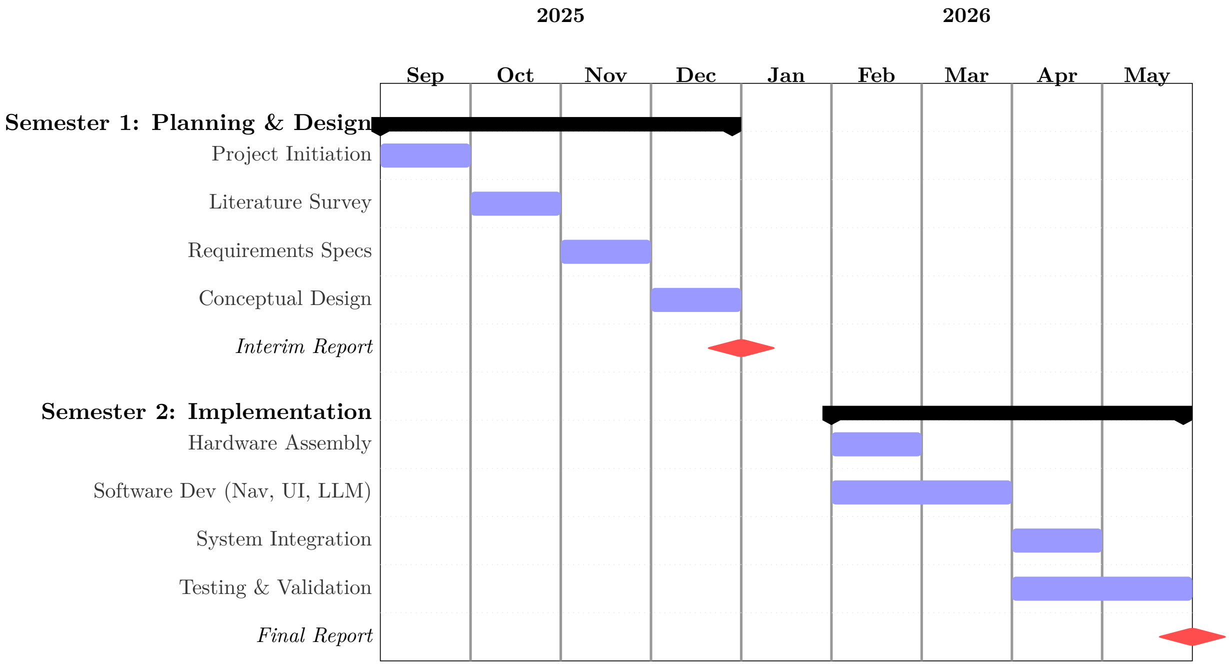

Figure 1: Project Gantt Chart (Sep 2025 - May 2026)

1.0.1 Work Breakdown Structure (WBS)

The detailed tasks and responsibilities are outlined in Table 1.

Table 1: Work Breakdown Structure (WBS)

Phase	Task Description	Deliverables	Assigned To
Phase 1	Project Initiation	Problem Statement	Both
(Sep 2025)	Methodology Definition	Methodology Section	Both
Phase 2	Literature Review	Literature Survey	Both
(Oct 2025)	Research Gap Analysis	Gap Identification	Both
Phase 3	Requirements Analysis	Requirements Doc	Both
(Nov 2025)	Constraints Analysis	Eng. Constraints	Both
Phase 4	Conceptual Design	Design Alternatives	Both
(Dec 2025)	Functional Decomposition	System Architecture	Both
Phase 5	Implementation: Hardware	Assembled Robot	Almoulla
(Feb-Mar)	Implementation: Navigation	Autonomous Nav	Almoulla
	Implementation: UI	Clinician/Patient GUI	Abdulmunim
	Implementation: LLM	Voice Interaction	Almoulla
Phase 6	Testing & Finalization	Integrated System	Both
(Apr-May)	Final Report Writing	Final Thesis	Both

2

Literature Review

This chapter surveys consumer, industrial, and healthcare robotics; documents benefits and challenges of AMRs in hospitals; analyzes core technologies (navigation, RF localization, SLAM, LLMs); and motivates the platform choices and project needs.

2 Literature Review

2.1 Motivation and State of the Field

The integration of robotics and artificial intelligence into healthcare is advancing rapidly, driven by challenges such as aging populations, rising healthcare costs, and medical staff shortages. Researchers have developed various robotic solutions, including telepresence robots for remote consultations, autonomous robots for logistics, and socially assistive robots (SARs) for patient engagement. While each type has proven useful, they often function in isolation, addressing only part of patients' and clinicians' overall needs. The goal is to show that although the components for such a system already exist, combining them into a single, user-friendly, and autonomous platform remains an underexplored area with high potential, especially for improving healthcare accessibility in regions like Oman.

Telepresence robots, often described as a mobile screen, are widely used for remote medical consultations. Systems such as InTouch Health RP-VITA and the VGo robot have been used successfully in "telestroke" care, allowing neurologists to assess patients remotely, reduce diagnosis time, and improve outcomes [5]. These systems support remote rounds, specialist consultations, and family visits, cutting travel costs and reducing infection risks. However, studies also note several drawbacks. A review by Kristoffersson et al. found that telepresence robots, though functional, often feel impersonal and difficult to control [6]. Clinicians report high cognitive load when navigating cluttered hospital spaces. Patients, especially the elderly and children, also find these robots less engaging than in-person visits due to limited physical embodiment and lack of expressive cues [7]. This "presence gap" limits their effectiveness beyond basic video conferencing, reducing their ability to offer comfort and reassurance during care.

This literature review builds a comprehensive case for an integrated, autonomous healthcare robotics platform. It begins by examining success stories from consumer and industrial robotics, demonstrating that autonomous navigation technology has reached maturity and widespread adoption. The review then analyzes healthcare-specific applications, including telepresence, socially assistive robots, and hospital logistics systems, alongside their underlying technical components such as navigation systems, mobile platforms, conversational AI, and user interfaces. By systematically evaluating these technologies and their implementations, this survey identifies a critical research gap: the absence of a unified system that synthesizes clinical telepresence, social interaction, and autonomous mobility into a single platform.

2.1.1 Success Stories: Consumer, Industrial, and Quadruped Robotics

1. **Consumer Market: The Robotic Vacuum Revolution.**
   The most successful deployment of autonomous mobile robots to the general public has been robotic vacuum cleaners, which dominate the consumer robotics market without serious competition from traditional cleaning methods. The global robotic vacuum cleaner market has experienced explosive growth, with projections indicating expansion from $9.07 billion in 2024 to $31.79 billion by 2033, representing a compound annual growth rate (CAGR) of 16.97% [8]. Global shipments reached 15.35 million units in the first half of recent years, demonstrating a 33% year-over-year growth [9].

Figure 2: Modern robotic vacuum cleaners utilize SLAM and localization technologies to autonomously navigate residential spaces, representing the most successful consumer robotics application to date.

The key to their success lies in the sophisticated implementation of SLAM (Simultaneous Localization and Mapping) and localization technologies. Modern robotic vacuum cleaners employ LiDAR sensors (Light Detection and Ranging), visual SLAM (VSLAM), and advanced mapping algorithms to create real-time maps of residential environments, enabling efficient path planning and obstacle avoidance [10]. This technology has proven so reliable that robotic vacuums have achieved nearly 40% of the service robot market share [11], demonstrating that autonomous navigation, when properly implemented, is mature enough for widespread consumer adoption.

2. **Industrial Market: Amazon's Warehouse Robotics Dominance.**
In the industrial sector, the most successful large-scale application of AMRs (Autonomous Mobile Robots) has been warehouse automation, with Amazon leading the field. Amazon reports more than 750,000 to over one million robots across its fulfillment network, including drive units, mobile manipulators, and next-generation platforms such as Pegasus and Xanthus [12, 13]. These fleets demonstrate reliable fleet management, safe human–robot collaboration, and 24/7 uptime in highly dynamic environments.

Figure 3: Amazon Pegasus drive unit used in high-throughput fulfillment centers.

3. **Quadruped Robotics: Unitree Go2 as a Robust Mobility Platform.**
State-of-the-art quadruped robots such as Unitree Go2 demonstrate robust mobility on unpredictable terrain using reinforcement learning for locomotion policy optimization, advanced multi-sensor perception (3D LiDAR + depth/RGB cameras), and high-performance

onboard compute for real-time planning and control. These capabilities enable operation on stairs, slopes, and uneven surfaces, with compelling applications in inspection, security, and emergency response [14, 15]. Although not our primary target platform, Unitree's capabilities illustrate complementary directions for rugged environments using RL and advanced multi-sensor perception, and inspire future extensions.

Figure 4: Unitree Go2: quadruped platform featuring RL locomotion, multi-sensor perception, and high-performance compute.

2.1.2 Benefits of Autonomous Mobile Robots in Healthcare Settings

Recent literature and deployment data reveal multiple evidence-based benefits of AMRs in hospital environments:

Infection Control and Reduced Human Contact. AMRs significantly minimize human-to-human contact, reducing cross-contamination risks a critical advantage demonstrated during the COVID-19 pandemic [16, 17]. Autonomous delivery robots eliminate the need for multiple staff members to handle potentially contaminated materials, reducing occupational exposure and conserving personal protective equipment. UV-C (ultraviolet-C) disinfection robots have been shown to achieve 99.9% pathogen kill rates on environmental surfaces, helping reduce hospital-acquired infections (HAIs) [18].

Wayfinding and Visitor Guidance. Intelligent hospital guidance robots provide accurate navigation assistance to visitors and patients, reducing confusion and staff burden [19]. These robots can escort visitors to patient rooms, provide multilingual directions, and answer frequently asked questions about hospital facilities, allowing staff to focus on clinical care.

Material Transport and Delivery. AMRs excel at routine delivery tasks including medications, meals, linens, laboratory specimens, and medical supplies. This automation frees nursing staff from non-clinical transport duties, allowing them to spend more time on direct patient care. Studies indicate that delivery robots can reduce staff time spent on logistics by up to 30% [20].

Psychological Support and Patient Morale. Research consistently demonstrates that pediatric patients exhibit positive emotional responses to robot interactions [21]. Socially interactive robots reduce anxiety, provide distraction during painful procedures, and improve treatment compliance in children. Studies show that patients perceive robotic companions as non-judgemental, reducing psychological barriers to seeking help or expressing concerns [22].

Autonomous Disinfection and Cleaning. Cleaning and disinfection robots can be equipped with UV-C light systems at relatively low cost, providing continuous environmental sanitation [16]. These robots operate during off-peak hours without human supervision, ensuring consistent hygiene standards while minimizing disruption to clinical workflows.

Telepresence and Virtual Presence. AMRs with integrated telepresence capabilities enable remote consultations, family visits, and specialist rounds without requiring travel [5]. This technology proved invaluable during pandemic restrictions and continues to provide value for rural healthcare access and specialist consultations.

Waste Collection and Handling. Some advanced AMR systems are deployed for biohazardous waste collection and disposal, reducing staff exposure to potentially infectious materials [20]. In China and other Asian markets, robots have been successfully implemented for collecting and transporting medical waste from patient rooms to centralized disposal areas, demonstrating significant improvements in operational efficiency and staff safety.

2.1.3 Challenges in Hospital AMR Deployment

Despite the significant benefits, several challenges remain:

Cost and Return on Investment. Hospital-grade AMRs represent substantial capital investments, typically ranging from $50,000 to $150,000 per unit depending on capabilities. Healthcare administrators require clear Return on Investment (ROI) demonstration through labor savings, efficiency gains, and improved outcomes to justify procurement [20].

System Integration Complexity. Integrating AMRs with existing hospital information systems (HIS), electronic health records (EHR), elevator controls, automatic doors, and staff communication systems presents significant technical challenges. Interoperability standards are still evolving, requiring custom integration work for each deployment [17].

Safety, Privacy, and Trust. Regulatory compliance (HIPAA, GDPR), patient data security, and physical safety in crowded environments remain critical concerns. Healthcare facilities must ensure that robots can safely navigate around patients, visitors, and medical equipment while maintaining patient privacy and data security [23]. Building trust among staff and patients requires careful change management and demonstration of reliability.

Regulatory and Liability Issues. Medical device regulations, liability frameworks for autonomous systems, and insurance considerations create additional barriers to widespread adoption. Healthcare organizations must navigate complex approval processes and establish clear accountability frameworks for robot operations.

2.1.4 Overview of Existing Robots in Healthcare

The hospital robotics market has matured significantly, with several manufacturers deploying autonomous mobile robots at scale. Table 2 presents a comprehensive comparison of leading hospital robot platforms currently deployed in eastern markets. Note that prices are approximate estimates and can vary significantly depending on region, configuration, volume, and additional features.

Rank	Manufacturer & Model	Est. Deployments	Est. Price (USD)	Key Hospital Applications	Features & Notes
1	Pudu Robotics – PuduBot 2 [24]	15,000–20,000 units (5,000 healthcare)	$12,000	Medication/meal delivery, linen transport, waste collection	Multi-floor navigation (elevator integration), VSLAM+LiDAR, 40 kg payload, 6–8 hr battery. Deployed in 100+ hospitals globally (China, Singapore, U.S.); customizable trays, infection-resistant coating.
2	Keenon Robotics – M2 Medical Robot [25]	8,000–10,000 units (healthcare-focused)	$47,500	UV-C disinfection, medication/lab sample delivery, patient room cleaning	UV sterilization (99.9% pathogen kill rate), 40 kg payload, voice interaction, 10 hr runtime. Used in 100+ Chinese hospitals during COVID; exported to U.S./Europe (HIPAA-compliant).
3	Pudu Robotics – PUDU CC1 [26]	5,000–7,000 units (healthcare + hospitality)	$16,800	Floor cleaning, disinfection, material transport	Autonomous scrubbing (1,500 m^2/hr), self-charging, obstacle avoidance. Popular in Chinese/U.S. hospitals for large-scale sanitation; integrates with PuduBot for hybrid tasks.
4	Keenon Robotics – Peanut Medical [27]	3,000–5,000 units	$12,700	Medication/meal delivery, patient guidance, telepresence	Compact (59 cm wide), multi-modal navigation, 30 kg payload. Deployed in Asia-Pacific hospitals; supports remote doctor-patient interaction via screen.

Table 2: Competitive analysis of leading hospital AMR platforms by global deployment and capabilities.

Figure 5: Pudu Robotics PuduBot 2

Figure 6: Keenon M2 Medical Robot

Figure 7: Pudu Robotics PUDU CC1

Figure 8: Keenon Peanut Medical

In parallel, the field of Socially Assistive Robotics (SAR) has focused on the psycho-social aspects of patient care. These robots are designed not to perform physical tasks but to engage, encourage, and comfort patients through social interaction. For pediatric patients, robots like NAO and PARO (a therapeutic baby seal robot) have been shown to significantly reduce pain perception and anxiety during painful medical procedures such as vaccinations and chemotherapy [21]. The robot acts as a coach and a distractor, creating a more positive and cooperative atmosphere. For elderly populations, SARs have been explored for companionship to combat loneliness, provide cognitive stimulation for dementia patients, and deliver medication reminders to improve treatment adherence [28]. These studies confirm that a robot's ability to interact socially is a powerful tool for improving patient well-being and clinical outcomes. The primary limitation of SARs, however, is that they are typically specialized, non-mobile or semi-autonomous platforms that lack the high-fidelity telepresence capabilities needed for direct clinical consultation.

2.2 Technical Building Blocks

This section deconstructs the essential technologies that form the foundation of an advanced autonomous healthcare robot. It examines navigation systems including Radio Frequency (RF) localization and SLAM-based sensor fusion, mobile robot platform specifications and selection criteria, Conversational AI and Large Language Models (LLMs) for intelligent interaction, and User Interface (UI) and Human-Robot Interaction (HRI) design paradigms for clinical integration.

2.2.1 Navigation System

The challenge of manual navigation highlighted in telepresence research is being addressed by autonomous mobile robots (AMRs). Modern AMRs employ a combination of SLAM, path planning, and obstacle avoidance to navigate complex environments without human intervention. The core navigation stack typically includes Radio Frequency (RF) Localization Technologies and SLAM-Based Sensor Fusion Techniques.

Radio Frequency (RF) Localization Technologies Radio frequency (RF) refers to electromagnetic waves (3 kHz–300 GHz) used for wireless communication. RF-based localization is essential where GPS is unreliable, such as indoors or through obstacles. Technologies like Wi-Fi, RFID, Bluetooth Low Energy (BLE), ZigBee, and Ultra-Wideband (UWB) are used, each selected based on coverage, scalability, battery life, accuracy, and cost. UWB, for example, enables multi-node scalability and multi-year tag battery life, but at higher cost (e.g., up to $20,000 for a 5000 m^2 facility). Accuracy ranges from about 20 cm to over 1 m, with performance degrading in cluttered or metallic environments [29]. The general structure of these systems is to use static reference sensor locations, or anchors, within an environment, along with beacons or tags fixed to the objects with desired tracking. The location of the tracked object can be estimated using readings from the anchors. Mathematically, this estimation is commonly called multilateration (also called trilateration) when using distance measurements. Figure 9 shows an example UWB anchor–tag geometry, while figure 10 illustrates the multilateration technique.

Figure 9: UWB anchor–tag geometry for ranging and multilateration-based localization in indoor environments.

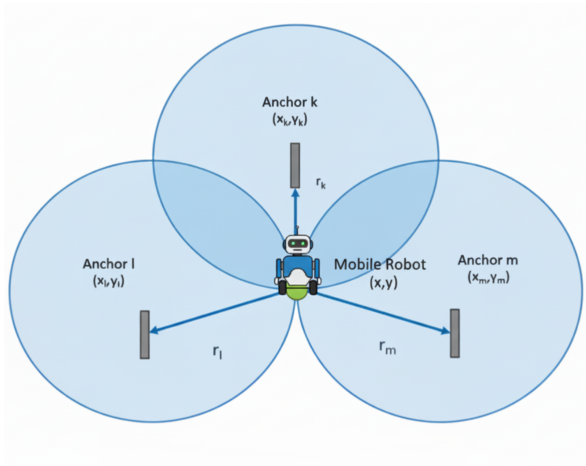

Figure 10: Illustration of multilateration technique. Each known anchor forms a circle with radius equal to its distance to the robot, and the robot's estimated position is where these circles intersect.

Angle of Arrival (AoA) is a localization technique that determines the direction or angle from which a radio signal is received. An antenna array senses time differences across elements and applies signal-processing algorithms such as MUSIC or ESPRIT to convert those differences into an angle. AoA (often via BLE 5.1+ or UWB Direction Finding) estimates a target's angle using a beacon or tag and a robot-mounted antenna array, reducing reliance on dense static infrastructure. BLE-based tags offer excellent battery life (years). Limitations include line-of-sight sensitivity and variable accuracy; hybrid AoA + RSSI filtering can improve precision, achieving mean errors on the order of 12 cm in simulation even in cluttered settings [30]. AoA could be used to implement a solution that does not depend on localization or positioning infrastructure in the environment, instead having the robot home in on a beacon worn by the user. This would be similar to how some Bluetooth trackers work for finding lost items. The robot would use the angle of arrival data to determine the direction to move towards the beacon signal, allowing it to autonomously navigate to the user's location. This approach could be particularly useful in dynamic environments like hospitals where fixed infrastructure may not be feasible. Figure 11 illustrates a proposed setup where a Bluetooth homing controller on the robot uses AoA data from a wearable beacon to guide navigation towards the target. Multiple anchors could be used to implement waypoints following navigation paths through the environment.

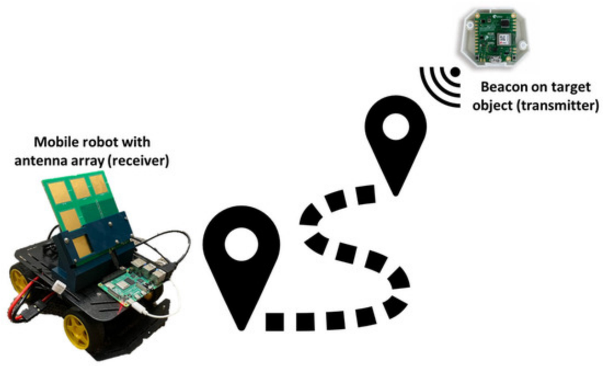

Figure 11: Configuration of proposed setup of Bluetooth homing controller.

SLAM-Based Sensor Fusion Techniques Simultaneous Localization and Mapping (SLAM) is a critical technology for autonomous navigation, enabling robots to build maps of unknown environments while simultaneously keeping track of their own location. Its effectiveness hinges on robust sensor fusion techniques that integrate data from various sources such as camera(s), IMU, wheel odometry, or RF ranging . Beyond external RF systems, autonomous navigation heavily relies on SLAM with multi-sensor fusion for robustness. A common stack fuses 2D LiDAR with an IMU and wheel odometry. IMU and odometry constrain motion, correcting LiDAR motion distortion and mitigating slip-induced drift. Studies report substantial gains when integrating IMU and wheel odometry, for example, approximately 27% improvement in rotation error and greater than 67% reduction in position error relative to LiDAR only SLAM, at the expense of added system complexity and compute [31].

VSLAM (Visual Simultaneous Localization and Mapping) is a method that uses cameras to help robots figure out where they are and build maps of their surroundings at the same time. VSLAM uses cameras like monocular (single), stereo (two), or RGB-D (depth-sensing) for positioning and mapping. It provides low-cost, lightweight hardware and better scene understanding. Challenges include poor performance in texture-less or repetitive areas, and scale issues in monocular systems (e.g., early ORB-SLAM 2.0). Modern approaches prefer hybrid Visual–Inertial–Ranging–LiDAR (VIRAL) SLAM for improved reliability in complex environments [32]. Figure 12 shows the combination of visual/laser data with inertial and odometric inputs for stable estimation. Tools like Google Cartographer build 3D maps in real-time, using LiDAR, IMU, and odometry.

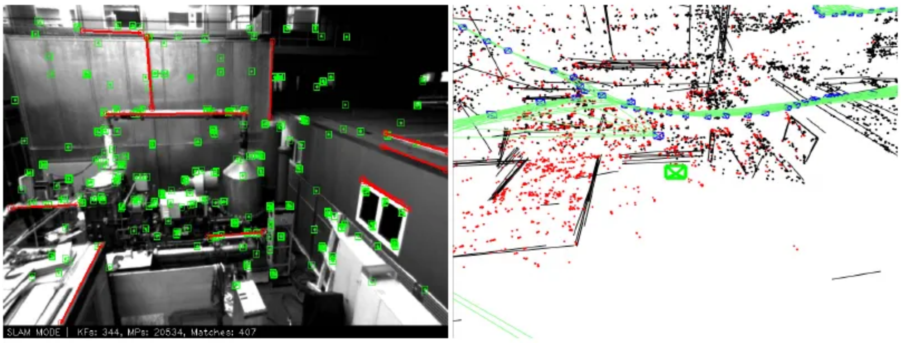

Figure 12: Illustration of SLAM fusion: visual/laser perception integrated with inertial and odometric constraints for robust state estimation.

Navigation Systems Across Hospital Robotics Platforms. Table 3 summarizes the navigation technologies and sensor configurations employed by leading hospital robotics platforms mentioned in table 2.

Robot Model	**Primary Localization**	**Mapping Technology**	**Key Sensors**
PuduBot 2 [24]	SLAM (LiDAR-based)	Real-time 2D/3D mapping	2D LiDAR, depth camera, IMU
M2 Medical Robot [25]	SLAM (LiDAR-based)	Real-time 2D mapping	2D LiDAR, RGB camera, encoders
PUDU CC1 [26]	SLAM (LiDAR-based)	Real-time 2D mapping	2D LiDAR, ultrasonic sensors, encoders
Peanut Medical [27]	Hybrid (SLAM + RF)	Real-time 2D mapping	LiDAR, depth sensors, optional UWB

Table 3: Navigation systems and sensor technologies across leading hospital robotics platforms.

2.2.2 Mobile Robot Platform specifications

The selection of an appropriate mobile robot platform is a critical decision that directly impacts system reliability, development time, and operational costs. For a healthcare telepresence robot operating in hospital corridors and patient rooms, the platform must satisfy several essential technical and operational requirements while remaining within practical budget constraints.

Platform Requirements. The system requires a wheeled mobile robot platform with the following capabilities:

- **Autonomous docking and charging** with high-level API commands (`dock()`, `return_to_charge()`)
- **ROS/ROS 2 compatibility** for navigation packages (Nav2, SLAM Toolbox, AMCL)
- **Payload capacity** of 5–10 kg for telepresence screen, onboard computer, and camera system
- **Indoor navigation sensors** including LiDAR (2D or 3D) and depth cameras
- **Safety and reliability** features (cliff detection, bumpers, emergency stop)

Project Constraints. Budget constraints significantly influence platform selection. The target budget for the mobile base, charging dock, and core sensors is approximately $1,000–$1,500 USD, recognizing that higher-cost industrial platforms ($5,000+) exceed the scope of an educational research project. Additional constraints include global availability (shipping to

Oman), vendor support and documentation quality, and community resources for troubleshooting and development.

Platform Comparison. Table 4 presents a comparative analysis of commercially available mobile robot platforms.

Platform	Price (USD)	Dock	ROS	Payload	Notes
Budget Tier ($500–$1,000)					
iRobot Create 3 [33]	$549	Yes	ROS 2	9 kg	Roomba-based, proven docking, ideal for education
Yahboom ROSMASTER X3 [34]	$699	DIY	ROS 1	5–8 kg	Mecanum wheels, requires custom docking solution
Elephant Robotics MyAGV 2023 [35]	$899	DIY	ROS 1	8 kg	Mecanum wheels, 2D SLAM ready, Python/ROS compatible
Mid Tier ($1,000–$2,000)					
SLAMTEC Apollo [36]	$1,200	Yes	ROS 1/2	50 kg	Same platform as Pudu/Keenon hospital robots
TurtleBot 4 Standard [37]	$1,400	Yes	ROS 2	9 kg	Research-grade, Nav2 pre-configured, Lidar included
Husarion ROSbot 2R [38]	$1,600	Optional	ROS 2	10 kg	Mecanum wheels, hospital navigation demos available
AgileX LIMO Pro [39]	$1,799	Optional	ROS 2	10 kg	4-mode steering, Jetson Nano, research platform

Table 4: Comparative analysis of mobile robot platforms for healthcare telepresence applications.

2.2.3 Conversational AI and Large Language Models (LLMs)

The field of large language models (LLMs) has witnessed transformative advancements, establishing them as a core technology for conversational AI. An LLM is a sophisticated artificial intelligence system trained on vast datasets of text and code, enabling it to understand, generate, and interact with human language in a nuanced manner. These models have rapidly transitioned from research prototypes to industrial-scale technologies, resulting in a proliferation of models with distinct strengths and commercial terms.

The LLM landscape is broadly divided into two categories: open-source and proprietary. Open-source models, such as Meta's Llama series and the DeepSeek series, offer significant flexibility, allowing for deep customization and fine-tuning to meet specific application needs, often at a lower cost [40]. However, they may require more complex setup and can sometimes lag behind proprietary counterparts in raw performance. In contrast, privately-owned models like OpenAI's GPT series and Google's Gemini series typically offer state-of-the-art performance and are adept at handling highly complex tasks. This performance comes with trade-offs, including higher operational costs and potential data privacy concerns, which are critical considerations in healthcare applications [41]. The selection of an LLM depends on a careful balance between performance requirements, budget, customizability, and data governance policies. Table 5 provides a comparative overview of leading LLM options.

Model (Version)	Org.	License	Max Context (tokens)	Input API Cost ($/M)	Output API Cost ($/M)
OpenAI GPT-5.2 [42]	OpenAI	Proprietary	128,000	$1.75	$14.00
Google Gemini 3 Flash [43]	Google	Proprietary	1,000,000	$0.50	$3.00
Google Gemini 2.5 Pro [43]	Google	Proprietary	1,000,000	$1.25–$2.50	$10.00–$15.00
Google Med-Gemma 2B	Google	Open Weight (Apache 2.0)	8,192	$0.00 (Self-hosted)	$0.00 (Self-hosted)
Meta Llama 4 Maverick [44]	Meta	Llama Community License	10,000,000	$0.19–$0.49	$0.19–$0.49
DeepSeek V3.1 [45]	DeepSeek	MIT (Code) / DeepSeek Model	128,000	$0.14	$0.28

Table 5: Comparative analysis of leading Large Language Models with December 2025 pricing and specifications [42, 43, 46, 44, 45].

LLM Discussion and Cost Considerations. Table 5 summarizes key characteristics of leading LLM options as of December 2025. OpenAI's GPT-5.2 offers strong performance with a 128k token context window at $1.75/$14.00 per million input/output tokens. Google's Gemini 3 Flash provides exceptional value with a 1M token context and competitive pricing at $0.50/$3.00, with free tier available for development (up to 500 requests per day). Google's Gemini 2.5 Pro offers higher capabilities for complex tasks at $1.25–$2.50 input and $10.00–$15.00 output per million tokens. Google's Med-Gemma represents a family of specialized open-weight models (available in 2B, 7B, and 27B parameter variants) fine-tuned specifically on medical literature, clinical guidelines, and healthcare datasets [46]. While Med-Gemma demonstrates state-of-the-art performance on medical benchmarks (MedQA, MedMCQA, PubMedQA) and would be highly relevant for advanced clinical decision support or medical question answering, it is considered overkill for the basic conversational interactions required in this project (patient guidance, wayfinding, simple information queries). Meta's Llama 4 Maverick stands out with an unprecedented 10M token context window and industry-leading cost efficiency ($0.19–$0.49 per million tokens), making it highly suitable for long-context healthcare applications. DeepSeek V3.1 offers the most economical option at $0.14/$0.28 per million tokens, particularly beneficial for cost-sensitive deployments. For this project's conversational requirements (patient guidance, wayfinding, simple information queries), either Gemini 3 Flash (free tier) or DeepSeek V3.1 (lowest cost) would be appropriate choices, with Llama 4 Maverick offering future-proof long-context capabilities if advanced document processing becomes necessary.

2.2.4 User Interface and Human-Robot Interaction

For autonomous healthcare robots operating in clinical environments, the user interface (UI) and human-robot interaction (HRI) design directly impacts adoption, user satisfaction, and clinical workflow integration. The interface must balance accessibility for diverse user groups (clinicians, administrative staff, patients) with robust control mechanisms suitable for dynamic hospital environments. Common UI paradigms employed in hospital robotics include:

Onboard Touchscreen Interface. A dedicated touchscreen mounted on the robot body provides direct, localized control and real-time status feedback. Common in systems like Pudu Robotics' PuduBot 2 and Keenon Robotics' M2 Medical Robot, these interfaces typically run

Android OS and display navigation status, task queues, and emergency controls. Advantages include intuitive interaction without external devices and immediate feedback. Limitations include reliance on the robot's physical proximity and potential hygiene concerns in sterile environments [25].

Mobile Application (iOS/Android). Native mobile applications running on clinicians' and administrators' smartphones or tablets enable remote operation and fleet monitoring from anywhere within hospital networks. This approach is ubiquitous across commercial systems (PuduBot 2, M2, Peanut Medical) and supports features such as route customization, real-time tracking, task scheduling, and status notifications. Mobile apps offer flexibility and integrate seamlessly with existing hospital workflows, though they require network connectivity and depend on user device management [24].

Web-Based Dashboard. Cloud-hosted or on-premises web portals provide centralized fleet management, historical analytics, and administrative controls for hospital operators and IT administrators. Systems like Keenon Robotics and Pudu Robotics include web dashboards for route optimization, performance metrics, and multi-robot coordination. Web interfaces support remote monitoring across multiple locations and facilitate integration with hospital information systems (HIS). Trade-offs include network dependency, data security considerations, and potential latency in real-time control [26].

Voice Control and Natural Language Interfaces. Voice-based interaction leveraging conversational AI and LLMs (as discussed in Section 2.2.3) represents an emerging paradigm for hands-free operation in clinical settings. Systems like Keenon's M2 support voice commands, allowing clinicians to interact with robots while maintaining hygiene and workflow continuity. Voice interfaces reduce cognitive load and support more natural, intuitive interactions, but require robust speech recognition, noise handling (hospital environments are loud), and contextual understanding [41].

Hospital Integration via APIs. Integration with existing hospital IT infrastructure through RESTful APIs or DICOM/HL7 standards allows robots to receive task assignments directly from electronic health records (EHR) or hospital management systems. This approach minimizes manual input and ensures tasks are coordinated with clinical workflows. Examples include systems capable of receiving delivery requests directly from hospital pharmacies or labs [26].

Robot UI Usage Across Hospital Platforms. Table 6 summarizes the UI and interaction modalities employed by leading hospital robotics platforms, illustrating the diversity of approaches and trade-offs in current systems.

Robot Model	**Onboard Screen**	**Mobile App**	**Web Dashboard**	**Voice/AI Interface**
PuduBot 2 [24]	Android tablet	iOS/Android	Yes	Limited
M2 Medical Robot [25]	Android display	iOS/Android	Yes	Yes (voice control)
PUDU CC1 [26]	Android panel	iOS/Android	Yes	No
Peanut Medical [27]	Optional	iOS/Android	Yes	Limited

Table 6: UI and interaction modalities across leading hospital robotics platforms.

2.3 Conclusion

The literature presents a clear picture:

1. **Telepresence robots** are effective for remote diagnosis but are limited by manual control and an impersonal user experience.
2. **Socially assistive robots** have proven their value in patient engagement and comfort but lack clinical telepresence and advanced mobility.
3. **Autonomous mobile robots** have mastered navigation in hospitals but are used for logistics, not patient interaction.
4. **Conversational AI** offers intelligent dialogue but lacks the physical embodiment needed for meaningful healthcare encounters.
5. **Consumer and industrial robotics** have demonstrated that autonomous navigation technology is mature, reliable, and cost-effective for real-world deployment, as evidenced by millions of robotic vacuum cleaners in homes and over one million AMRs in Amazon warehouses.
6. **Current hospital AMRs** are predominantly focused on material transport and disinfection, with limited integration of social interaction, advanced telepresence, and conversational AI capabilities.

The critical research gap lies at the intersection of these domains. There is a demonstrable need for a single, integrated platform that combines the **clinical access** of a telepresence robot, the **patient-centric engagement** of a SAR, the **operational independence** of an autonomous navigator, and the **interactive intelligence** of modern conversational AI. The proven success of autonomous navigation in consumer (robotic vacuums) and industrial (Amazon warehouses) applications demonstrates that the core technology is mature and ready for advanced healthcare applications.

This project directly addresses this gap by designing a holistic system that synthesizes these strengths, creating a smart healthcare robot capable of not only connecting doctors and patients but also navigating its environment independently and engaging with patients in a helpful and empathetic manner. By learning from the successes of consumer robotics, warehouse automation, and existing hospital AMR deployments, this project aims to advance the state-of-the-art in healthcare robotics toward truly integrated, intelligent, and patient-centered care.

3

System Conceptual Design

This chapter presents the high-level system architecture of the smart healthcare robot, including the overall system overview, design alternatives, and justification for the selected hardware and software configuration.

3 System Conceptual Design

3.1 Introduction: System Overview

The proposed smart healthcare robot is a mobile, autonomous platform designed to assist hospital staff in routine, non-critical tasks while providing patients with an intuitive human–robot interaction (HRI) interface. The robot operates primarily in indoor clinical environments such as hospital wards, corridors, and waiting areas. Its main roles include autonomous navigation, safe obstacle avoidance, point-to-point delivery of small items, basic patient interaction via conversational AI, and remote telepresence capabilities for clinical staff.

From a functional perspective, the robot must be able to localize itself in the hospital environment, plan safe paths between waypoints, and react to dynamic obstacles such as people, trolleys, and doors. It should expose a user-friendly interface through a tablet and voice interaction for nurses, doctors, and patients to issue tasks or receive information. The system is conceived as a modular platform so that sensors, software modules, and interaction modalities can be extended in future iterations without major redesign.

The overall system architecture follows a layered approach consisting of:

- **Mobile Base Layer**: Providing locomotion, payload mounting, and physical robustness
- **Computing and Control Layer**: High-level computer running ROS 2 and microcontroller for real-time motor control
- **Navigation and Perception Layer**: Sensors for localization, mapping, and obstacle detection
- **Intelligence Layer**: Large Language Model (LLM) for natural language understanding and HRI
- **Communication and Interface Layer**: Tablet UI, web dashboard, and voice interaction
- **Power Management Layer**: Battery system with automatic docking capabilities

This conceptual design aims to balance technical performance, patient safety, modularity, and cost, providing a realistic path from prototype to potential deployment in Omani hospitals.

3.2 Design Alternatives and Selection

To arrive at a suitable system architecture for the smart healthcare robot, three main design alternatives were systematically evaluated. The evaluation focuses on robot platform approach, computing architecture, sensor configuration, and AI integration capabilities.

3.2.1 Alternative A: Commercial Pre-Built Platform

This alternative leverages an existing commercial robot platform designed for healthcare or service applications, such as the TurtleBot 4, Robotnik RB-1, or similar ROS-compatible mobile bases.

System Architecture

- **Mobile Base**: Commercial differential-drive or mecanum-wheel platform with integrated motors, encoders, and basic safety sensors
- **Computing**: Pre-integrated single-board computer (typically Raspberry Pi 4/5 or Intel NUC) running ROS 2

- **Navigation**: Factory-installed 2D LiDAR (RPLiDAR A1/A2 or Hokuyo), IMU, and wheel odometry
- **Perception**: Optional RGB-D camera (Intel RealSense or similar) for 3D perception
- **Intelligence**: Cloud-based LLM integration via API calls (OpenAI GPT-4, Anthropic Claude, or Google Gemini)
- **Interface**: Standard ROS 2 interfaces, web-based dashboard, optional Android tablet mount
- **Power**: Integrated battery pack (typically Li-ion) with proprietary or standard docking station

Advantages

- Significantly reduced development time with pre-tested hardware integration
- Professional mechanical design with proven reliability and safety features
- Comprehensive documentation, tutorials, and community support
- Pre-calibrated sensors and validated ROS 2 navigation stack
- Commercial warranty and technical support availability
- Established supply chain for replacement parts

Limitations

- Higher initial cost (typically $3,000–$8,000 depending on configuration)
- Limited customization of mechanical structure and form factor
- Vendor lock-in for certain components and software stacks
- May include unnecessary features increasing system complexity
- Longer lead times for procurement and shipping to Oman
- Less hands-on learning experience for mechanical and low-level integration

3.2.2 Alternative B: Custom-Built Modular Platform

This alternative involves designing and fabricating the entire robot from individual components, including mechanical structure, motor selection, electronics integration, and software development.

System Architecture

- **Mobile Base**: Custom-designed aluminum extrusion frame with selected DC motors (e.g., 12V with encoders) and 4WD mecanum or omni wheels
- **Computing**: Raspberry Pi 5 (4GB) as main computer running Ubuntu + ROS 2, ESP32 for low-level motor control
- **Navigation**: Separately procured RPLiDAR A1, MPU6050 IMU, and encoder-based odometry

- **Perception**: USB webcam or Pi Camera Module 3 for vision, ultrasonic sensors for safety
- **Intelligence**: On-device quantized LLM (e.g., Gemini Nano, LLaMA 3.2 3B) or hybrid cloud approach
- **Interface**: Custom Android tablet application, Flask-based web dashboard
- **Power**: 12V $LiFePO_4$ battery (10–20Ah) with custom BMS, DC–DC converters, and DIY docking station

Advantages

- Maximum design flexibility and customization for hospital-specific requirements
- Significantly lower cost ($800–$1,500 estimated total)
- Complete understanding of all system components and integration points
- Ability to iterate quickly on mechanical and electrical design
- Educational value through hands-on experience with robotics fundamentals
- Easy sourcing of replacement parts from local suppliers or online marketplaces
- Full control over software stack and no vendor dependencies

Limitations

- Substantially longer development time (6–9 months for initial prototype)
- Higher technical risk requiring expertise in mechanical design, electronics, and software
- No warranty or commercial support; troubleshooting relies on team knowledge
- Requires multiple design iterations to achieve reliability comparable to commercial systems
- Need for specialized tools and facilities (3D printers, soldering equipment, testing instruments)
- Potential safety concerns requiring rigorous testing and validation
- Limited payload capacity and robustness compared to engineered commercial platforms

3.2.3 Alternative C: Hybrid Pre-Built Base with Custom Intelligence

This alternative combines a commercial robot base platform with custom-developed upper layers, particularly focusing on advanced AI-driven HRI and task management capabilities.

System Architecture

- **Mobile Base**: Commercial platform (e.g., TurtleBot 4, Yahboom ROS-MASTER X3) providing proven navigation
- **Computing**: Base platform computer (Raspberry Pi 4/5) plus optional edge AI accelerator (Google Coral TPU or NVIDIA Jetson Nano) for on-device inference
- **Navigation**: Pre-integrated LiDAR, IMU, and odometry from base platform
- **Perception**: Added RGB-D camera and microphone array for enhanced HRI

- **Intelligence**: Custom integration of Google Gemini 3 flash via API for conversational AI, with fallback to on-device models
- **Interface**: Custom Android tablet application with voice interaction, hospital-specific task workflows, and real-time telemetry dashboard
- **Power**: Platform's integrated battery with custom power monitoring and management software

Advantages

- Balanced approach: reliable base platform with custom high-value features
- Faster time to functional prototype (3–4 months) compared to full custom build
- Focus development effort on AI/HRI innovation rather than low-level integration
- Commercial platform provides safety-certified mobility while allowing software customization
- Moderate cost ($2,000–$3,500) compared to fully commercial or fully custom
- Flexibility to swap AI models and interaction modalities without hardware redesign
- Proven navigation reliability combined with cutting-edge conversational AI

Limitations

- Still dependent on commercial base platform availability and lead times
- Less mechanical customization than Alternative B
- Requires cloud connectivity for full LLM capabilities, raising data privacy considerations
- Integration complexity between commercial ROS stack and custom AI modules
- Potential latency issues with cloud-based LLM requiring careful system design
- Limited ability to modify low-level control and sensor fusion algorithms

3.3 System Selection

3.3.1 Robot Platform Selection Using AHP

To systematically select the most appropriate robot platform alternative, we employ the Analytic Hierarchy Process (AHP), a structured decision-making methodology. Four key criteria are identified: **Cost**, **Development Time**, **Customizability**, and **Reliability**.

Table 7: Robot Platform Feature Comparison

Feature	A: Commercial	B: Custom-Built	C: Hybrid
Motors Integrated	Yes (pre-installed DC motors with encoders)	No (requires selection & integration)	Yes (commercial base motors)
Sensors Integrated	Yes (LiDAR, IMU, odometry pre-calibrated)	No (procure & calibrate separately)	Yes (base sensors included)
Computer Integrated	Yes (RPi 4/5 or NUC pre-configured)	No (requires setup & ROS installation)	Yes (base computer + optional upgrade)
LLM Integrated	No (requires custom integration)	No (requires custom integration)	No (custom Gemini 3 flash integration)
ROS 2 Compatibility	Yes (native support)	Yes (manual configuration)	Yes (native + custom nodes)
CE/EMC Standards	Yes (commercial certification)	No (requires testing)	Partial (base certified, additions not)
IP Rating	IP42–IP54 (varies by model)	None (custom design)	Base platform rating (IP42)
Safety Standards	ISO 13482 compliance typical	None (requires validation)	Base compliant, custom features TBD
Documentation	Comprehensive (manuals, ROS packages)	Community resources only	Base docs + custom development
Warranty/Support	1–2 years commercial warranty	None	Base warranty only

Platform Comparison Summary

Criteria Pairwise Comparison The pairwise comparison matrix for criteria (scale: 1 = equal importance, 3 = moderate importance, 5 = strong importance, 7 = very strong importance, 9 = extreme importance):

Table 8: Platform Criteria Pairwise Comparison Matrix

Criterion	Cost	Dev. Time	Customizability	Reliability
Cost	1	1/3	3	1/5
Development Time	3	1	5	1/3
Customizability	1/3	1/5	1	1/7
Reliability	5	3	7	1

Criteria Weights After normalization and calculating the priority vector:

- **Cost**: 0.15 (15%)
- **Development Time**: 0.30 (30%)
- **Customizability**: 0.08 (8%)
- **Reliability**: 0.47 (47%)

Consistency Ratio (CR) = 0.04 < 0.10, indicating acceptable consistency.

Alternative Scoring Each alternative is scored against each criterion (scale: 1–9, where 9 = best):

Table 9: Platform Alternative Scores by Criterion

Alternative	Cost	Dev. Time	Customizability	Reliability
A: Commercial	3	9	3	9
B: Custom-Built	9	3	9	5
C: Hybrid	6	7	6	7

Final AHP Scores Calculating weighted scores: Score = $\sum$(Criterion Weight$\times$Alternative Score)

Table 10: Final AHP Scores for Robot Platform

Alternative	Final Score
A: Commercial Pre-Built	7.11
B: Custom-Built Modular	5.55
C: Hybrid Platform	**6.76**

While Alternative A (Commercial) scores highest, **Alternative C (Hybrid)** is selected as it offers the best balance of reliability (47% weight) and development speed (30% weight) while maintaining sufficient customizability for healthcare-specific AI features. The modest score difference (7.11 vs 6.76) is acceptable given that Alternative C enables innovation in HRI–the project's primary differentiator–while leveraging proven navigation hardware.

3.3.2 Computing Platform Selection Using AHP

Three computing platforms are evaluated for the main onboard computer: Raspberry Pi 5, NVIDIA Jetson Nano, and Intel NUC.

Computing Platform Criteria

- **Processing Power**: CPU/GPU performance for ROS 2 and AI workloads
- **Power Consumption**: Battery life impact
- **Cost**: Hardware acquisition cost
- **Ecosystem Support**: ROS 2 compatibility, community, documentation

Table 11: Computing Platform AHP Summary

Platform	Power (40%)	Consumption (25%)	Cost (20%)	Ecosystem (15%)	Final Score
Raspberry Pi 5	6	9	9	9	**7.65**
Jetson Nano	8	7	5	7	7.05
Intel NUC	9	4	3	8	6.55

Selected: Raspberry Pi 5–Optimal balance of sufficient processing power for ROS 2 Nav2, excellent power efficiency (critical for battery operation), low cost, and extensive ROS community support.

3.3.3 Navigation Approach Selection Using AHP

Three navigation approaches are compared: LiDAR-based SLAM, Visual SLAM, and Hybrid SLAM.

Navigation Criteria

- **Accuracy**: Localization precision in hospital corridors
- **Robustness**: Performance in dynamic/challenging environments
- **Computational Cost**: Processing requirements
- **Sensor Cost**: Hardware acquisition cost

Table 12: Navigation Approach AHP Summary

Approach	Accuracy (35%)	Robustness (35%)	Comp. Cost (15%)	Sensor Cost (15%)	Final
LiDAR SLAM	9	8	7	6	**7.95**
Visual SLAM	7	6	5	9	6.65
Hybrid SLAM	8	9	4	5	7.35

Selected: LiDAR-based SLAM (Nav2 + RPLIDAR A1)–Superior accuracy and robustness in structured indoor environments, well-established ROS 2 Nav2 integration, acceptable sensor cost.

3.3.4 Localization Sensor Selection Using AHP

Three primary localization sensor configurations are evaluated.

Localization Sensor Criteria

- **Precision**: Pose estimation accuracy
- **Update Rate**: Sensor refresh frequency
- **Integration Complexity**: Software and hardware setup difficulty
- **Cost**: Total sensor package cost

Table 13: Localization Sensor AHP Summary

Configuration	Precision (40%)	Update (20%)	Integration (20%)	Cost (20%)	Final
LiDAR + Odom + IMU	9	8	8	7	**8.20**
Vision + Odom + IMU	7	7	6	8	7.00
UWB + Odom + IMU	6	9	5	4	6.10

Selected: LiDAR + Odometry + IMU–Best precision for structured hospital environments, proven ROS 2 sensor fusion with robot_localization package, extensive Nav2 support.

3.3.5 LLM Selection Using AHP

For the conversational AI component, three state-of-the-art LLM options are evaluated: OpenAI GPT-5.2, OpenAI GPT-4 Turbo, and Google Gemini 3 flash (announced December 2024).

LLM Evaluation Criteria Four criteria are defined:

- **API Familiarity**: Team experience with the API and SDK
- **Performance**: Response quality, reasoning capability, and medical domain knowledge
- **Latency**: Speed of response generation for real-time interaction
- **Cost**: API pricing per token (input/output)

Table 14: LLM Criteria Pairwise Comparison Matrix

Criterion	**API Familiarity**	**Performance**	**Latency**	**Cost**
API Familiarity	1	1/3	3	5
Performance	3	1	5	7
Latency	1/3	1/5	1	3
Cost	1/5	1/7	1/3	1

Criteria Pairwise Comparison for LLM

LLM Criteria Weights After normalization:

- **API Familiarity**: 0.26 (26%)
- **Performance**: 0.54 (54%)
- **Latency**: 0.14 (14%)
- **Cost**: 0.06 (6%)

Consistency Ratio (CR) = 0.05 < 0.10, acceptable.

Table 15: LLM Alternative Scores by Criterion

oprule extbfLLM	**API Familiarity**	**Performance**	**Latency**	**Cost**
GPT-5.2	7	9	6	5
GPT-4 Turbo	5	9	7	6
Gemini 3 flash	9	8	9	9

LLM Alternative Scoring Scoring rationale:

- **GPT-5.2**: Excellent performance, moderate familiarity, good but not fastest latency, moderate cost
- **GPT-4 Turbo**: Excellent performance with strong safety features, less team familiarity, good latency, moderate cost
- **Gemini 3 flash**: High team familiarity with Google AI APIs, excellent performance for size, extremely low latency optimized for real-time use, very cost-effective

Table 16: Final AHP Scores for LLM Selection

LLM	**Final Score**
GPT-5.2	7.62
GPT-4 Turbo	7.54
Gemini 3 flash	**8.44**

Final LLM AHP Scores **Google Gemini 3 flash** is selected based on its superior AHP score. Key advantages include:

- Team's extensive experience with Google Cloud Platform and Gemini API
- Optimized for low-latency real-time conversational AI (critical for natural HRI)
- Multimodal capabilities (text, image, audio) enabling future vision-based features
- Native integration with Google Cloud services used in the project
- Significantly lower cost enables extensive testing and development iterations
- Announced Flash variant (Dec 2024) specifically designed for speed-critical applications

3.3.6 System Selection Summary

Table 17: Final System Component Selection Summary

Component	**Selected Option**	**AHP Score**
Robot Platform	Hybrid (Commercial Base + Custom AI)	6.76
Computing Platform	Raspberry Pi 5 (4GB)	7.65
Navigation Approach	LiDAR-based SLAM (Nav2)	7.95
Localization Sensors	LiDAR + Odometry + IMU	8.20
LLM	Google Gemini 3 flash	8.44

3.4 Selected System Architecture

Based on the AHP analysis, the final system architecture combines:

- **Robot Platform**: TurtleBot 4 or Yahboom ROS-MASTER X3 (commercial base)
- **Main Computer**: Raspberry Pi 5 (4GB) running Ubuntu 22.04 + ROS 2 Humble
- **Navigation**: Integrated RPLiDAR A1, MPU6050 IMU, wheel encoders
- **Perception**: Pi Camera Module 3 + USB microphone array
- **AI/LLM**: Google Gemini 3 flash via Cloud API
- **Interface**: Custom Android tablet app with voice interaction
- **Power**: Platform's Li-ion battery with automatic docking

3.5 Functional Decomposition

3.5.1 System Level 0: Black Box View

At the highest abstraction level, the smart healthcare robot can be represented as a single system with external interfaces. Figure 13 summarizes the Level 0 functional view, grouping the main inputs, outputs, and primary functions.

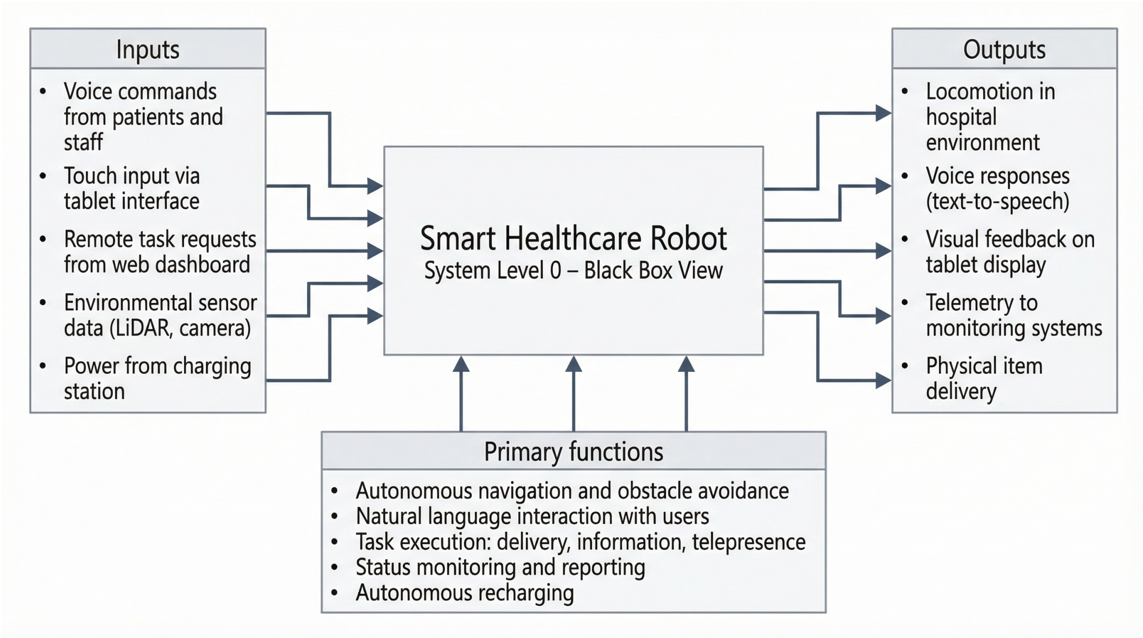

Figure 13: System Level 0 black-box view of the smart healthcare robot, showing external inputs, outputs, and primary functions.

For clarity, the main interfaces can be described textually as follows:

- **Inputs**:
  - Voice commands from patients/staff
  - Touch input via tablet interface
  - Remote commands from web dashboard
  - Environmental sensor data (LiDAR scans, camera images)
  - Power from charging station
- **Outputs**:
  - Locomotion (movement through hospital environment)
  - Voice responses (text-to-speech)
  - Visual feedback (tablet display)
  - Telemetry data to monitoring systems
  - Physical item delivery
- **Primary Functions**:
  - Autonomous navigation and obstacle avoidance
  - Natural language interaction with users

– Task execution (delivery, information provision, telepresence)
– Status monitoring and reporting
– Autonomous recharging

3.5.2 System Level 1: Subsystem Decomposition

The system is decomposed into seven major subsystems, each with defined responsibilities. Figure 14 presents the Level 1 functional decomposition and the main interfaces between subsystems.

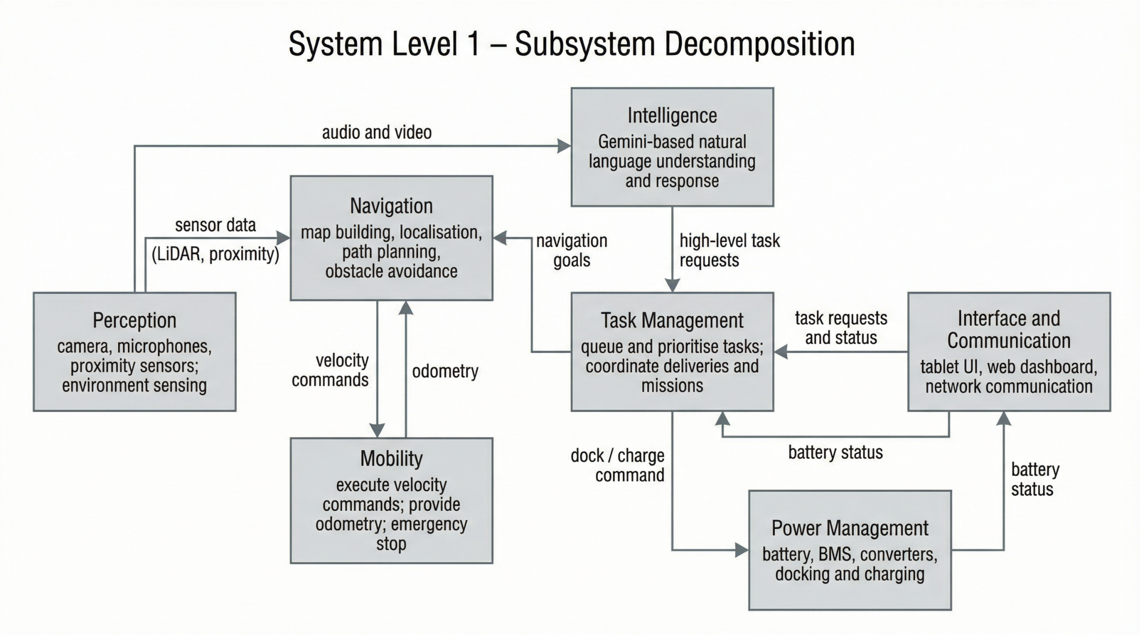

Figure 14: System Level 1 subsystem decomposition of the smart healthcare robot, highlighting the seven functional subsystems and key data/control flows.

The subsystems are defined textually as follows:

1. Mobility Subsystem

- **Components**: 4WD mecanum base, DC motors, motor drivers, wheel encoders
- **Functions**: Translate velocity commands to wheel motion, provide odometry feedback, execute emergency stops
- **Interfaces**: Receives velocity commands from Navigation Subsystem, sends encoder data

2. Navigation Subsystem

- **Components**: RPLiDAR A1, MPU6050 IMU, ROS 2 Nav2 stack (SLAM, localization, path planning)
- **Functions**: Build and update maps, localize robot pose, plan collision-free paths, detect obstacles
- **Interfaces**: Receives goal poses from Task Manager, sends velocity commands to Mobility, provides pose estimates to all subsystems

3. **Perception Subsystem**
    - **Components**: Pi Camera Module 3, microphone array, ultrasonic/ToF sensors
    - **Functions**: Capture visual data for teleoperation and docking, record audio for voice interaction, detect close-range obstacles
    - **Interfaces**: Sends camera frames to Interface Subsystem and Intelligence Subsystem, audio to Intelligence Subsystem, proximity alerts to Navigation Subsystem

4. **Intelligence Subsystem**
    - **Components**: Gemini 3 flash API client, speech-to-text, text-to-speech, dialogue manager
    - **Functions**: Process natural language queries, generate contextual responses, manage conversation state, translate between voice and text
    - **Interfaces**: Receives audio from Perception, text commands from Interface, sends responses to Interface, task requests to Task Manager

5. **Task Management Subsystem**
    - **Components**: Task scheduler, state machine, goal queue, hospital map database
    - **Functions**: Queue and prioritize tasks, track task execution status, coordinate between subsystems, manage waypoints and delivery locations
    - **Interfaces**: Receives task requests from Intelligence and Interface, sends navigation goals to Navigation, status updates to Interface

6. **Interface and Communication Subsystem**
    - **Components**: Android tablet app, Flask web dashboard, ROS 2 bridge, Wi-Fi module
    - **Functions**: Display status and controls, enable touch/voice input, provide remote monitoring, transmit telemetry, receive remote commands
    - **Interfaces**: Bidirectional communication with all subsystems, external network connection to web dashboard and cloud services

7. **Power Management Subsystem**
    - **Components**: 12V Li-ion battery, BMS, DC–DC converters, docking contacts, charging station
    - **Functions**: Distribute power to all subsystems, monitor battery status, initiate autonomous docking when low, manage charging process
    - **Interfaces**: Provides power to all subsystems, sends battery status to Task Management and Interface, receives docking commands from Task Management

Subsystem Interaction Flow As summarized in Figure 14, the robot operates in a closed loop of sensing, decision-making, and actuation. The Perception subsystem collects sensor data and forwards it to Navigation for localisation and path planning, while also sending audio and video streams to the Intelligence subsystem. Intelligence interprets user intent and generates high-level task requests, which are queued and coordinated by Task Management. Task Management issues navigation goals to Navigation, which computes velocity commands for the Mobility subsystem and receives odometry feedback in return. Interface and Communication exposes task requests, status information, and telemetry to users and remote dashboards, while Power Management responds to dock/charge commands and reports battery status to both Task Management and the interface layer. This interaction pattern ensures that environmental sensing, user intent, motion control, and power management remain synchronised during hospital missions.

3.6 Conclusion

This chapter presented a systematic approach to the conceptual design of the smart healthcare robot. Through AHP-based evaluation of three platform alternatives (commercial, custom-built, and hybrid), the hybrid approach was selected as it balances reliability with customizability for healthcare-specific AI features. A second AHP analysis selected Google Gemini 3 flash as the optimal LLM for conversational AI due to team familiarity, low latency, and cost-effectiveness.

The resulting system architecture is decomposed into seven functional subsystems (Mobility, Navigation, Perception, Intelligence, Task Management, Interface, and Power Management), each with clearly defined responsibilities and interfaces. This modular design enables parallel development, facilitates testing and validation, and provides a clear path for future enhancements such as manipulation capabilities or multi-robot coordination.

The selected architecture provides a solid foundation for the detailed design and implementation phases that follow, ensuring the robot meets the project objectives of safe autonomous operation, natural human–robot interaction, and practical deployment in Omani hospitals.

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